source: curriculum stage-1 winner + representative nonzero feedback gains; failed stage-3 score -159.45255762990192 ignored - edges drawn from the COMPILED network object (build_network.py, LAMINATED architecture), representative knee column x46/side | network size (live): 46 MN + 46x(3) sensory per side, +12 rhythm/pattern cells + 5 load-sharing INs per side; both sides + 8 shared = 410 neurons, 376 inputs

SUPRASPINAL (inputs)

DRIVE

MLR speed cmd

POSTURE

tonic

BAL family (COM PD, IMU trunk, lateral -> ankle/hip/abd/trunk MNs in the full net) omitted here

0.8

1.6

RHYTHM GENERATOR (per side; right shown)

2.2

InE

RG-E

NaP

RG = persistent-Na cond. bursters (Deng 2022, Shinohara 2025, Rybak 2024): intrinsic h-gate burst termination, FIXED tau_h (= period knob) + weak mutual excitation G_W (escape mode; speed neuromodulation). ADAP retired 2026-09-16

V3

IN-laminated mutual inhibition (Shevtsova/Deng A6: RG-E->InE->RG-F, RG-F->InF->RG-E; no direct HC<->HC synapses)

InF

RG-F

c1

HEEL

TOE

NaP

v11 contact mechanosensors (S2W trigger + late-stance prolongation; same extensor central pathway as Ib)

c1_r -> INHIBIT RG-F_l (g=2.4); V3_r -> EXCITE InE_l (g=1.0) + contra extensor MNs (P2a)

PATTERN FORMATION (4 phase-window cells, laminated cross-reciprocal via PF_INs)

PF_IN_E

PF-E1

PF-E2

PF-F1

PF-F2

PF_IN_F

KINH cond.

IN-laminated PF cross-reciprocal (Shevtsova/Deng A6; no direct PF<->PF)

v6 swing quad-suppression IN

MOTOR CIRCUIT - knee column (representative; x46 muscle columns/side)

POSTURE (+ per-MN POST_i standing bias)

PF->MN x W_PF_MN(group) (F2 = late-swing prep)

MN knee-ext

RC

Renshaw: MN->RC 1.0, RC->MN g=0.50 mutual RC<->RC (distinct pools)

RC

MN knee-flx

same complete MN--muscle--afferent motif repeats for the other 44 hip, ankle, frontal-plane, and trunk muscle pools

RG-E stance gate g=1.0 (loaded extensor groups only; x5 groups)

LBIN stance-Ib IN (RG layer, Dominguez 2020; load prolongs stance)

IB-EXC

LBIN

PF-F1 phase gate (0.5)

per-muscle afferent encoders (x92)

stance load sharing (reflex reversal)

Ia

II

Ib

IaIN

IaIN

Ia

II

Ib

Ia homonymous exc
II length exc
Ib autogenic inh

Ia reciprocal inhibition via IaIN: Ia->IaIN (g=0.60), PF-F1-gated; IaIN->MN g=0.40; RC->IaIN = recurrent disinhibition

per-muscle afferent -> central (same-group exc); extensor Ib(+II) -> PF_E / RG_E / InE (Rybak SF-E2, force feedback prolongs stance); flexor Ia+II -> PF_F / RG_F / InF (SF-E1, length feedback triggers swing); flexor Ia also inhibits the CONTRALATERAL RG-F (SF-E1, interleg)

NEURO-MUSCULAR INTERFACE (conversion maps; circles)

pure-signal encoders; gains are phase- and speed-gated presynaptically in runner.py

clip(Ldot/0.6)

-> Ia

(L-Lmid)/Lhalf

-> II

clip(F/Fmax)

-> Ib

a=clip(V/5mV, 0,1)

MN -> activation

MuJoCo supplies muscle length, velocity, and force; the complete interface repeats for all 92 muscles

clip(Ldot/0.6)

-> Ia

(L-Lmid)/Lhalf

-> II

clip(F/Fmax)

-> Ib

a=clip(V/5mV, 0,1)

MN -> activation

knee-ext muscle (MuJoCo Hill F-L-V, rigid tendon, Fmax = OpenSim)

knee-flx muscle (same MuJoCo model)

neuron

excitatory

inhibitory

conversion map (continuous <-> neural)

all drawn cells are IMPLEMENTED in build_network.py (edges read back from the compiled network)

Deng et al./Nourse 2023 Table A6 (rat hindlimb SNS) - reference values:
HC->IN 2.75 uS exc | IN->HC 2.75 uS inh (RG+PF mutual inhibition IS LAMINATED through dedicated INs; no direct HC<->HC synapses, no mutual excitation)
RGHC->PFHC 0.10 uS exc | PF->MN 1.5-4.9 uS exc (per joint)
PF->Ia 0.50 uS exc (phase gate) | IaIN->MN 2.0 uS inh | IaIN<->IaIN 0.5 inh
MN->RC 0.50 uS exc | RC->MN 0.50 uS inh | RC<->RC 0.5 inh | RC->IaIN 0.5 inh
IbIN->MN 0.59 uS EXCITATORY (positive force feedback) | PF->Ib 2.0 uS shunt
MN->muscle: act = 1/(1+exp(s(x0-V)))+y0, s=0.153, x0=-70 mV (Fig 6B)
feedback: MuJoCo muscle TENSION formatted as Ia and Ib input (no II)